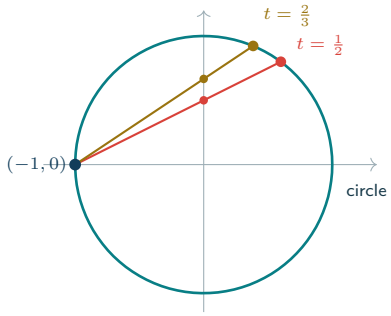
PROJECTIVE GEOMETRY LADDER 7 OF 9

Pythagorean triples from the projective line

A rational slope gives a rational point on the circle, and clearing denominators gives the triple.

1 / THE RULE AND THE PICTURE

Every line through $(-1,0)$ with rational slope t meets the circle again at a rational point



Putting $y = t(x+1)$ into $x^2 + y^2 = 1$ gives $(1+t^2)x^2 + 2t^2x + t^2 - 1 = 0$, a quadratic with one root $x = -1$, so the other root is rational too, by Bézout and the product of the roots:

$$(x, y) = \left(\tfrac{1-t^2}{1+t^2},\ \tfrac{2t}{1+t^2}\right)$$

With $t = \frac{n}{m}$ this is $\frac{m^2-n^2}{m^2+n^2}, \frac{2mn}{m^2+n^2}$, the legs over the hypotenuse of a right-angled triangle.

2 / HOMOGENEOUS COORDINATES

Clearing denominators turns the parameter $(m : n)$ into a point of the conic

(m,n)	m^2-n^2	$2mn$	m^2+n^2	point $(X : Y : Z)$
$(2,1)$	3	4	5	$(3:4:5)$
$(3,2)$	5	12	13	$(5:12:13)$
$(4,1)$	15	8	17	$(15:8:17)$
$(4,3)$	7	24	25	$(7:24:25)$
$(5,2)$	21	20	29	$(21:20:29)$
$(5,4)$	9	40	41	$(9:40:41)$

The map $(m : n) \mapsto (m^2 - n^2 : 2mn : m^2 + n^2)$ sends the projective line to the conic $X^2 + Y^2 = Z^2$, with $m = 0$ giving the base point $(-1 : 0 : 1)$, and the complex parameters $n = \pm im$ giving $Z = 0$ and the two circular points $(1 : \pm i : 0)$.
A Pythagorean triple is then a rational point of the conic written with coprime integer coordinates, and it is primitive when $\gcd(a, b, c) = 1$.

3 / EVERY PRIMITIVE TRIPLE APPEARS ONCE

Coprime $m > n$ of opposite parity

Every primitive triple has the form $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$ for a unique pair $m > n > 0$ with $\gcd(m, n) = 1$ and $m - n$ odd, and swapping the legs is the only freedom. Two pairs can give the same hypotenuse, as $(8, 1)$ gives $(63, 16, 65)$ and $(7, 4)$ gives $(33, 56, 65)$, and a pair of equal parity gives twice a smaller triple, as $(3, 1)$ gives $(8, 6, 10) = 2(4, 3, 5)$.

4 / THE FAMILY TREE OF TRIPLES

Three matrices preserve the conic and generate all primitive triples from $(3, 4, 5)$

The matrices $\begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}$ and $\begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$ send $(3, 4, 5)$ to $(5, 12, 13)$, $(21, 20, 29)$ and $(15, 8, 17)$, and each satisfies $M^T Q M = Q$ for $Q = \mathrm{diag}(1, 1, -1)$, so it is a projective map carrying the conic $X^2 + Y^2 = Z^2$ to itself. Berggren showed in 1934 that repeating them reaches every primitive triple exactly once.

5 / HOW MANY TRIPLES

About $N/2\pi$ primitive triples have hypotenuse up to N

Counting with the formula gives 16 primitive triples with $c \le 100$ against $100/2\pi = 15.9$, and 158 with $c \le 1000$ against 159.2, the first being $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$ and $(20, 21, 29)$.

6 / TRY THESE

Use (m, n)

(a) Find the triple for $(m, n) = (4, 3)$: $(7, 24, 25)$, and check $49 + 576 = 625$.
(b) Which pair gives $(20, 21, 29)$? $(5, 2)$, since $25 - 4 = 21$ and $2 \cdot 10 = 20$.
(c) Why does $(6, 8, 10)$ not come from a coprime pair of opposite parity? Because it is twice $(3, 4, 5)$.

ASIDE / A RATIONAL CIRCLE

The same line trick parametrises the rational points of every conic that has one, which is why conics have an easy theory of rational points and cubics do not.

LINKS TO OTHER SHEETS

The number theory ladder has the Pythagorean triples through Gaussian integers, and the next sheet shows what replaces this method for cubics.

Convention. (m, n) coprime with $m > n$ and $m - n$ odd gives a primitive triple; $Q = \mathrm{diag}(1, 1, -1)$ is the quadratic form of the conic; Berggren, 1934.

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